

SCIENTIFIC DISCOVERIES, NOBEL PRIZES AND SCIENTISTS

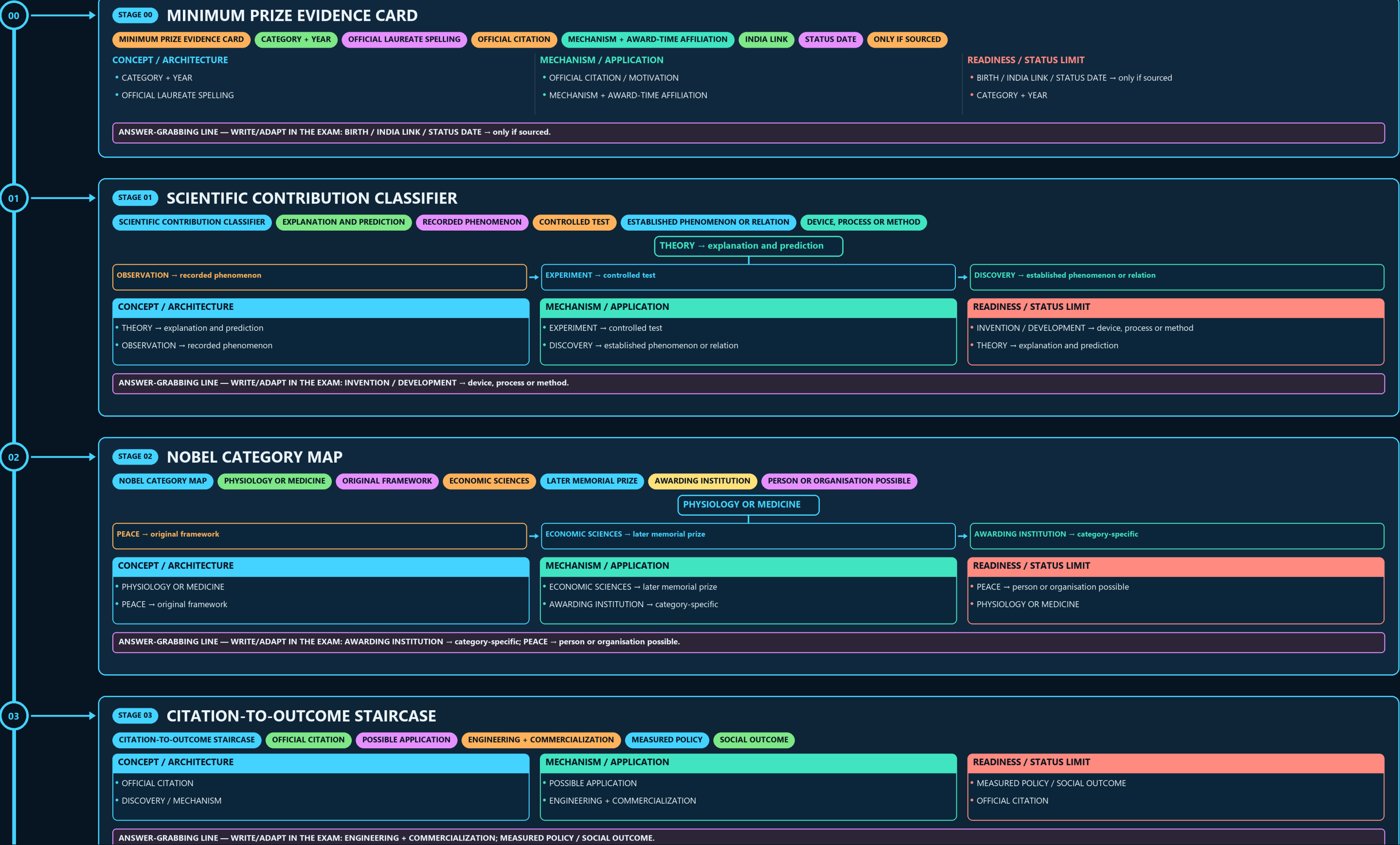
MINIMUM PRIZE EVIDENCE CARD → SCIENTIFIC CONTRIBUTION CLASSIFIER → NOBEL CATEGORY MAP → CITATION-TO-OUTCOME STAIRCASE → ATTRIBUTION AND CHRONOLOGY RAIL → PHYSICS 2025 QUANTUM-CIRCUIT MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STAGE 03

CITATION-TO-OUTCOME STAIRCASE

CITATION-TO-OUTCOME STAIRCASE

OFFICIAL CITATION

POSSIBLE APPLICATION

ENGINEERING + COMMERCIALIZATION

MEASURED POLICY

SOCIAL OUTCOME

CONCEPT / ARCHITECTURE

• PHYSIOLOGY OR MEDICINE

• PEACE → original framework

MECHANISM / APPLICATION

• ECONOMIC SCIENCES → later memorial prize

• AWARDING INSTITUTION → category-specific

READINESS / STATUS LIMIT

• PEACE → person or organisation possible

• PHYSIOLOGY OR MEDICINE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AWARDING INSTITUTION → category-specific; PEACE → person or organisation possible.

04

STAGE 04

ATTRIBUTION AND CHRONOLOGY RAIL

ATTRIBUTION AND CHRONOLOGY RAIL

PRIOR THEORY

LATER AND SEPARATE

AWARD YEAR

DISCOVERY YEAR AUTOMATICALLY

PRIOR THEORY → INSTRUMENT → EXPERIMENT

PUBLICATION → REPLICATION → AWARD

PATENT / PRODUCT / DEPLOYMENT → later and separate

LAUREATE != institution

AWARD YEAR != discovery year automatically

CONCEPT / ARCHITECTURE

• PRIOR THEORY → INSTRUMENT → EXPERIMENT

• PUBLICATION → REPLICATION → AWARD

MECHANISM / APPLICATION

• PATENT / PRODUCT / DEPLOYMENT → later and separate

• LAUREATE != institution

READINESS / STATUS LIMIT

• AWARD YEAR != discovery year automatically

• PRIOR THEORY → INSTRUMENT → EXPERIMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PATENT / PRODUCT / DEPLOYMENT → later and separate.

05

STAGE 05

PHYSICS 2025 QUANTUM-CIRCUIT MAP

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SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

COLLECTIVE MACROSCOPIC STATE

QUANTUM TUNNELLING

SPECIFIC ENERGY AMOUNTS

FAULT-TOLERANT QUANTUM COMPUTER

COLLECTIVE MACROSCOPIC STATE

BARRIER → quantum tunnelling

SPECIFIC ENERGY AMOUNTS → quantisation

CONCEPT / ARCHITECTURE

• SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

• COLLECTIVE MACROSCOPIC STATE

MECHANISM / APPLICATION

• BARRIER → quantum tunnelling

• SPECIFIC ENERGY AMOUNTS → quantisation

READINESS / STATUS LIMIT

• DEMONSTRATION != fault-tolerant quantum computer

• SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECIFIC ENERGY AMOUNTS → quantisation; DEMONSTRATION != fault-tolerant quantum computer.

06

STAGE 06

JOHN CLARKE CLUE RESOLVER

JOHN CLARKE CLUE RESOLVER

CAMBRIDGE, UNITED KINGDOM

UNIVERSITY OF CALIFORNIA, BERKELEY

PHYSICS 2025

ONE-THIRD WITH DEVORET AND MARTINIS

MATCH ALL CLUES

THEN IDENTIFY

BORN → Cambridge, United Kingdom

AFFILIATION → University of California, Berkeley

CATEGORY / YEAR → Physics 2025

SHARE → one-third with Devoret and Martinis

CONCEPT / ARCHITECTURE

• BORN → Cambridge, United Kingdom

• AFFILIATION → University of California, Berkeley

MECHANISM / APPLICATION

• CATEGORY / YEAR → Physics 2025

• SHARE → one-third with Devoret and Martinis

READINESS / STATUS LIMIT

• MATCH ALL CLUES → then identify

• BORN → Cambridge, United Kingdom

SCIENTIFIC DISCOVERIES, NOBEL PRIZES AND SCIENTISTS • TILE 2/4 • same master rows 2257-5491 of 10004 • continues below

- DEMONSTRATION != fault-tolerant quantum computer
- SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECIFIC ENERGY AMOUNTS → quantisation; DEMONSTRATION != fault-tolerant quantum computer.

STAGE 06 JOHN CLARKE CLUE RESOLVER

THEN IDENTIFY

BORN → Cambridge, United Kingdom

→ **SHARE** → one-third with Devoret and Martinis

MECHANISM / APPLICATION

- CATEGORY / YEAR → Physics 2025
- SHARE → one-third with Devoret and Martinis

- MATCH ALL CLUES → then identify
- BORN → Cambridge, United Kingdom

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SHARE — one-third with Devoret and Martinis; MATCH ALL CLUES — then identify.

STAGE 07 **CHEMISTRY 2025 MOF ARCHITECTURE**

TUNABLE FUNCTION

MECHANISM / APPLICATION

- POROUS CRYSTAL → cavities
- BUILDING-BLOCK CHANGE → tunable function

- CAPABILITY / APPLICATION != commercialization
- METAL IONS → nodes

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUILDING-BLOCK CHANGE → tunable function; CAPABILITY / APPLICATION != commercialization.

STAGE 08 MEDICINE 2025 TOLERANCE PATHWAY

CLINICAL TRIAL

OUTPUT / FAILURE BOUNDARY

- CLINICAL TRIAL != approved therapy
- CENTRAL TOLERANCE → thymic control layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SELF-TISSUE PROTECTION → mechanism; CLINICAL TRIAL != approved therapy.

STAGE 09 MICRORNA REGULATION RAIL

MUST REMEMBER

▶ **MUST REMEMBER:** Scientific-history questions separate observation, discovery, theory

MECHANISM / APPLICATION

- DISCOVERY != diagnostic or medicine
- MUST REMEMBER: Scientific-history questions separate observation, discovery, theory,...

- DNA --transcription--> mRNA microRNA → complementary mRNA binding
- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Scientific-history questions separate observation, discovery, theory..

STAGE 10 INDIA-LINKED SCIENTIST CARDS

RAMAKRISHNAN 2009

09

STAGE 09 MICRORNA REGULATION RAIL

MICRORNA REGULATION RAIL DNA --TRANSCRIPTION--> MRNA MICRORNA COMPLEMENTARY MRNA BINDING PROTEIN PRODUCTION INHIBITED MRNA DEGRADATION POSSIBLE ROUTE DIAGNOSTIC OR MEDICINE MUST REMEMBER



CONCEPT / ARCHITECTURE

- DNA --transcription--> mRNA microRNA → complementary mRNA binding
- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route

MECHANISM / APPLICATION

- DISCOVERY != diagnostic or medicine
- MUST REMEMBER: Scientific-history questions separate observation, discovery, theory,...

READINESS / STATUS LIMIT

- DNA --transcription--> mRNA microRNA → complementary mRNA binding
- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Scientific-history questions separate observation, discovery, theory,.

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STAGE 10 INDIA-LINKED SCIENTIST CARDS

INDIA-LINKED SCIENTIST CARDS RAMAN 1930 LIGHT SCATTERING RAMAN EFFECT KHORANA 1968 GENETIC CODE PROTEIN SYNTHESIS RAMAKRISHNAN 2009

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
RAMAN 1930 → light scattering / Raman effect	RAMAKRISHNAN 2009 → ribosome structure and function	CITATION != every later application
KHORANA 1968 → genetic code / protein synthesis	BIRTHPLACE != award-time affiliation	CLOSE DISTINCTION: Discovery is not invention, theory is not proof beyond revision, award...

CONCEPT / ARCHITECTURE

- RAMAN 1930 → light scattering / Raman effect
- KHORANA 1968 → genetic code / protein synthesis

MECHANISM / APPLICATION

- RAMAKRISHNAN 2009 → ribosome structure and function
- BIRTHPLACE != award-time affiliation

READINESS / STATUS LIMIT

- CITATION != every later application
- CLOSE DISTINCTION: Discovery is not invention, theory is not proof beyond revision, award...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: KHORANA 1968 → genetic code / protein synthesis.

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STAGE 11 PYQ AND FUTURE-STATUS FIREWALL

PYQ AND FUTURE-STATUS FIREWALL 2026 Q81 DIRECT BIOGRAPHICAL PRIZE ROUTE BLUE LED TOPIC 21 DIRECT SCIENCE OWNER TOPIC 26 ATTRIBUTION + CITATION + MATURITY SUPPORT 2025 OFFICIAL AWARDS



SYSTEM / DEVICE / INSTITUTION

- 2026 Q81 → direct biographical prize route
- BOSE-EINSTEIN / BLUE LED → Topic 21 direct science owner

DECISIVE TECHNICAL DISTINCTION

- TOPIC 26 → attribution + citation + maturity support
- 2025 OFFICIAL AWARDS → latest completed science anchors

STATUS / UNIT / EVIDENCE LIMIT

- 4 SEP 2026 → no invented 2026 laureates
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use Nobel Foundation or equally...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use Nobel Foundation or equally...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use Nobel Foundation or equally.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT USE LAUREATE EXAMPLES TO ILLUSTRATE LONG-HORIZON PUBLIC RESEARCH, INSTRUMENTATION MOBILITY AND THE PATH FROM BASIC SCIENCE TO STRATEGIC AVOID "NOBEL COUNT" NATIONALISM.

A STRONGER ANSWER EXAMINES RESEARCH FUNDING, UNIVERSITIES, LABORATORY AUTONOMY CONNECT TOPIC 24'S FUNDING ARCHITECTURE WITH THE SPECIALIST SCIENCE SCIENCE & TECHNOLOGY GS PAPER

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Use laureate examples to illustrate long-horizon public research, instrumentation, international
- mobility and the path from basic science to strategic technology.
- CAUTION: Avoid "Nobel count" nationalism.
- A stronger answer examines research funding, universities, laboratory autonomy, interdisciplinary teams and translation capacity.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Connect Topic 24's funding architecture with the specialist science file that explains the discovery.
- Subject: Science & Technology | Tier: Advanced | GS Paper: GS-III + Prelims.
- Purpose: Convert prize current affairs into deeper scientific reasoning without making optional enrichment necessary for objective-question coverage.
- Analytical lens CAUTION: Major prizes reveal how science moves from a counter-intuitive

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- mechanism to an experimental demonstration and then to a technology platform.
- Deeper distinctions | Distinction | Why it matters | [---|---] | Discovery vs invention | A discovery establishes a phenomenon
- an invention creates a device or process.
- Prize citations may recognise either kind of contribution.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.